

Computer System III

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国家级“一流本科”专业配套教材
浙江省首批普通本科高校“1411”重点建设教材
浙江省理工类专业建设计算机系列教材
计算机科学与技术

计算机系统贯通课程 实践教材 RISC-V架构

常瑞 中文博 吴磊 周正金 著



教学课件



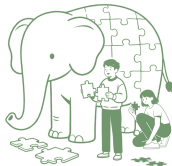
教学大纲



教学计划



配套资源



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Before the Class

Talk about this course

- What are you learn from computer system I and II?



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- Which software courses did you complete?



Talk about this course

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- How the hardware influence the software or program?



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 - Know not only what but also why.



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- Which software courses did you complete?
- How the hardware influence the software or program?
- What will be covered in our course and you should take away:
 - Understand the working principles of Computer.
 - Know not only what but also why.
 - Explore the tradeoffs of different designs and ideas.
 - Your own system. (How about it?)





Prepare for the Class

What's More

How I Prepared before the class.

- Textbooks



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- Course Website (<https://courses.zju.edu.cn/course/93687/>)



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- Reference
- Teaching Components
 - Lectures
 - Projects
 - Research



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 - Lectures
 - Projects
 - Research
- About me



Me

Rui Chang

- Associate Professor, College of CS, ZJU



Me

Rui Chang

- Associate Professor, College of CS, ZJU
- Research Interests:
 - Security for AI
 - System Security: ARM/RISC-V Architecture, Linux/Android Security, Hypervisor-based, Hardware-assisted
 - Agent AI System Security: Access Control, Multi-Agent Security
 - AI for Security
 - AI for Program Analysis
 - AI for Formal Verification



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- My website (<https://person.zju.edu.cn/changrui>)



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My style (according to the course content and the learners)

- Keep it simple and clear



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- More concerns about Agentic AI System
 - XPU Vulnerability
 - Memory Security
 - System Security



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- Keep it simple and clear
- Focus on the core concepts
- Try to help you more easily understand
- More concerns about Agentic AI System
 - XPU Vulnerability
 - Memory Security
 - System Security
- Warm-up for research



You

Some suggestions for you.

- Books
 - Computer Organization and Design The Hardware Software Interface [RISC-V Edition]
 - Operating System Concept
 - (some part of) Computer Architecture: A Quantitative Approach [6th edition]
 - Elephant Book by ZJU (2025)



You

Some suggestions for you.

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- Other ways
 - from class (lectures/labs/projects)
 - from github (<https://github.com/riscv/riscv-isa-manual>)
 - doing by yourself (of course, discussion with others)
 - optional extension



You should know:

Scoring

- Final examination(30%)



You should know:

Scoring

- Final examination(30%)
- Process assessment(70%)
 - Homework and Class Attendance(10%)
 - Projects(60%)
 - Lab 1 - BHT & BTB(10%)
 - Lab 2 - Cache design(10%)
 - Lab 3 - Virtual Memory(8%)
 - Lab 4 - User mode(8%)
 - Lab 5 - Page fault and fork system call(10%)
 - Project - TBD(14%)



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Important note

- Homework/Labs need to be submitted in time.



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Important note

- Homework/Labs need to be submitted in time.
- Late work penalized 10%/day.
- Any new try is strongly encouraged.



How will you contribute?

Thanks in advance.

- Know the real architecture
 - according to reading, thinking, discussion and doing
 - show your lab/project demo
 - literature reading about recent research



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Thanks in advance.

- Know the real architecture
 - according to reading, thinking, discussion and doing
 - show your lab/project demo
 - literature reading about recent research
- AT LEAST
 - submit assignments & lab reports
 - show up to final exam





Begin the Class

Contents

- 1. Fundamentals of computer design----- 1 week
- 2. Instruction-Level Parallelism (ILP)----- 2.5 weeks
- 3. Memory Hierarchy (Cache)----- 2.5 weeks
- 4. Memory Hierarchy (Main Memory)----- 4 weeks
- 5. File System----- 4 weeks
- 6. DLP, TLP and Summary----- 2 weeks



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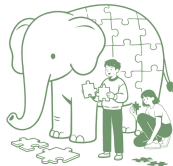


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